

Code Review: Fortune Credit Data Analysis

1. Fix the harvest timing calculation (highest priority)

In `app.py` (line 34), `Days_to_Harvest` is converted to text before being queried in SQLite. The timing comparisons therefore use text ordering instead of numeric ordering.

I reproduced these incorrect classifications:

- 212 days and 184 days are classified as 0–30 days.
- 90 days is classified as 90+ days.

The correct results for the supplied dataset are:

Days before harvest	Farmer loans	Repaid	Repayment rate
0–30	0	0	No observations
31–90	2	2	100%
Over 90	5	2	40%

Requested improvement: Keep day counts numeric, represent missing values as null, and create display labels separately. Apply the fix to both scripts, then regenerate the charts. Test boundary values such as 30, 31, 90, and 91 days.

2. Make the conclusions match the evidence

The README (line 14) claims significantly better repayment in the 31–90 day window compared with both earlier and later lending.

The dataset contains 10 mock loans, including seven farmer loans, and no farmer loans in the 0–30 day window. This cannot establish statistical significance, an optimal lending window, or a causal effect of timing.

Requested improvement: Describe the result as an illustrative observation. For example:

“In this mock dataset, both farmer loans issued 31–90 days before harvest were repaid, compared with two of five loans issued over 90 days before harvest. More representative data is needed to evaluate this pattern.”

Include sample sizes beside every rate. To investigate whether repayment aligns with harvest income, also collect loan due dates and actual repayment dates.

3. Correct the default-rate chart

At `app.py` (line 112), the chart receives both `Total_Loans` and `Default_Rate`. This mixes counts and percentages in a chart titled “Default Rate.”

Requested improvement: Plot only `Default_Rate`, clearly label the percentage scale, and show loan counts separately.

4. Define the repayment metrics

The calculation at `app.py` (line 79) divides defaults by all loans. The dataset includes a Late loan, so a segment can show 0% defaults while still having a repayment problem.

Requested improvement: Explain what “Defaulted” and “Late” mean, state the denominator, and display late loans separately. For real portfolio analysis, establish which loans have had sufficient time to reach an outcome.

5. Make the project reproducible

`clean_data.py` (line 3) requires Matplotlib, and its Excel export requires OpenPyXL. Neither is declared in `requirements.txt`; the Pipfile also has no packages.

The README’s clone command creates a directory with a trailing hyphen, but its `cd` command omits it.

Requested improvement: Declare all required dependencies, choose a consistent dependency setup, correct the installation instructions, and verify both entry points in a fresh environment.

6. Improve data handling and documentation

- Share cleaning and analysis functions between the two scripts so fixes stay consistent.
- Distinguish a farmer’s missing harvest date from a non-agricultural loan.
- Validate required columns, duplicate loan IDs, invalid dates, negative harvest gaps, and unexpected repayment statuses.
- Flag imputed loan amounts. Median imputation can change the mean, contrary to the claim in README (line 24).
- Update the README’s description of Matplotlib rendering: the current dashboard uses native Streamlit charts.

Suggested closing feedback

You have demonstrated a useful range of skills by combining Pandas, SQL, and Streamlit into a working analysis structure. Your next priority is analytical correctness: preserve numeric types, verify group assignments, show sample sizes, and keep conclusions within the evidence. Correct the timing calculation and chart, make the setup reproducible, and revise the findings before resubmitting.